

Fisher Information for Air Sonar Distance Estimation

ELEC3305 Tutorial / Lecture Notes

Craig T. Jin

April 28, 2026

Learning objectives

By the end of this tutorial, students should be able to:

1. explain Fisher Information as a measure of how much a signal tells us about an unknown parameter;
2. derive Fisher Information and the Cramér–Rao lower bound for a simple Gaussian-noise model;
3. apply these ideas to time-delay estimation using a matched filter;
4. explain why wide-band chirps improve range resolution in an air sonar system;
5. connect signal-to-noise ratio, bandwidth, sampling rate, pulse duration, and distance accuracy.

1 The AIR SONAR project: what are we estimating?

In the real-time AIR SONAR project, the laptop loudspeaker transmits a pulse or chirp, and the microphone records the echo from an object. The goal is to estimate the distance to the object.

The basic physical relationship is

$$d = \frac{c\tau}{2}, \quad (1)$$

where

- d is the distance to the reflecting object;
- c is the speed of sound in air;
- τ is the round-trip time delay between the transmitted pulse and the echo;
- the factor 2 appears because sound travels to the object and back.

A common approximation for the speed of sound is

$$c \approx 331.3 + 0.606T \quad \text{m/s}, \quad (2)$$

where T is the temperature in degrees Celsius. At $T = 20^\circ\text{C}$,

$$c \approx 343.4 \text{ m/s}. \quad (3)$$

For example, an object at $d = 1 \text{ m}$ produces a round-trip delay

$$\tau = \frac{2d}{c} \approx \frac{2}{343.4} \approx 5.82 \text{ ms}. \quad (4)$$

At a sampling rate of $f_s = 48 \text{ kHz}$, this corresponds to

$$N_\tau = f_s \tau \approx 48000 \times 0.00582 \approx 279 \text{ samples}. \quad (5)$$

Key idea

The sonar distance-estimation problem is mainly a *time-delay estimation* problem. Fisher Information tells us how accurately the delay, and therefore the distance, can be estimated from a noisy recorded signal.

2 Signal model for a single echo

A simplified continuous-time model for the received signal is

$$r(t) = As(t - \tau) + w(t), \quad (6)$$

where

- $s(t)$ is the transmitted pulse or chirp;
- A is the echo amplitude, which depends on loudspeaker strength, microphone gain, object reflectivity, and distance;
- τ is the unknown delay;
- $w(t)$ is additive noise.

In discrete time, sampled at f_s , we may write

$$r[n] = As[n - n_\tau] + w[n], \quad (7)$$

where $n_\tau \approx \tau f_s$.

For mathematical derivations it is convenient to keep the delay τ as a continuous parameter:

$$r[n] = As(nT_s - \tau) + w[n], \quad T_s = \frac{1}{f_s}. \quad (8)$$

3 Matched filtering and delay estimation

The matched filter computes a correlation between the received signal and the known transmitted pulse. In continuous time,

$$y(\lambda) = \int r(t)s(t - \lambda) dt. \quad (9)$$

If $r(t)$ contains a delayed copy of $s(t)$, then $y(\lambda)$ should have a peak near $\lambda = \tau$.

In discrete time, the cross-correlation is approximately

$$y[k] = \sum_n r[n]s^*[n - k]. \quad (10)$$

The estimated delay is often chosen as

$$\hat{k} = \arg \max_k |y[k]|, \quad \hat{\tau} = \frac{\hat{k}}{f_s}. \quad (11)$$

The estimated distance is therefore

$$\hat{d} = \frac{c\hat{\tau}}{2} = \frac{c\hat{k}}{2f_s}. \quad (12)$$

Key idea

The matched filter gives us a practical algorithm. Fisher Information gives us a theoretical limit: how well any unbiased algorithm could estimate the delay from the noisy data.

4 What is Fisher Information?

Suppose we observe data $\mathbf{x}$ whose probability density depends on an unknown parameter θ . The likelihood is

$$p(\mathbf{x}; \theta). \quad (13)$$

The Fisher Information is

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log p(\mathbf{x}; \theta) \right)^2 \right]. \quad (14)$$

The term

$$\frac{\partial}{\partial \theta} \log p(\mathbf{x}; \theta) \quad (15)$$

is called the *score*. It measures how sensitive the log-likelihood is to changes in θ .

Key idea

Large Fisher Information means the probability distribution changes sharply when the parameter changes. Therefore the data contain a lot of information about the parameter.

5 The Cramér–Rao lower bound

If $\hat{\theta}$ is an unbiased estimator of θ , then

$$\text{var}(\hat{\theta}) \geq \frac{1}{I(\theta)}. \quad (16)$$

This is the Cramér–Rao lower bound, often abbreviated CRLB.

For the sonar problem, if we estimate the delay τ , then

$$\text{var}(\hat{\tau}) \geq \frac{1}{I(\tau)}. \quad (17)$$

Since distance is $d = c\tau/2$, the distance estimate satisfies

$$\hat{d} = \frac{c\hat{\tau}}{2}. \quad (18)$$

Therefore

$$\text{var}(\hat{d}) = \left(\frac{c}{2}\right)^2 \text{var}(\hat{\tau}). \quad (19)$$

Hence

$$\boxed{\text{var}(\hat{d}) \geq \left(\frac{c}{2}\right)^2 \frac{1}{I(\tau)}}. \quad (20)$$

6 Fisher Information for a signal in Gaussian noise

Assume the observed discrete-time signal is

$$x[n] = \mu[n; \theta] + w[n], \quad n = 0, 1, \dots, N-1, \quad (21)$$

where $w[n]$ is independent Gaussian noise:

$$w[n] \sim \mathcal{N}(0, \sigma^2). \quad (22)$$

Then

$$p(\mathbf{x}; \theta) = \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x[n] - \mu[n; \theta])^2}{2\sigma^2} \right]. \quad (23)$$

The log-likelihood is

$$\log p(\mathbf{x}; \theta) = C - \frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - \mu[n; \theta])^2, \quad (24)$$

where C does not depend on θ .

Differentiating with respect to θ gives

$$\frac{\partial}{\partial \theta} \log p(\mathbf{x}; \theta) = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - \mu[n; \theta]) \frac{\partial \mu[n; \theta]}{\partial \theta}. \quad (25)$$

Taking the expected square of this score gives the Fisher Information. We will do this in detail explicitly.

Start by defining

$$\mu_\theta[n] = \frac{\partial \mu[n; \theta]}{\partial \theta}$$

and recall that

$$x[n] - \mu[n; \theta] = w[n].$$

Therefore the score can be written as

$$u(\mathbf{x}; \theta) = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} w[n] \mu_\theta[n].$$

Justification of the score re-expression From the signal model

$$x[n] = \mu[n; \theta] + w[n],$$

we can rearrange to obtain

$$x[n] - \mu[n; \theta] = w[n].$$

Substituting this into the expression for the score function,

$$\frac{\partial}{\partial \theta} \log p(\mathbf{x}; \theta) = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - \mu[n; \theta]) \frac{\partial \mu[n; \theta]}{\partial \theta},$$

gives

$$\frac{\partial}{\partial \theta} \log p(\mathbf{x}; \theta) = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} w[n] \frac{\partial \mu[n; \theta]}{\partial \theta}.$$

For notational convenience, define

$$\mu_\theta[n] = \frac{\partial \mu[n; \theta]}{\partial \theta}.$$

Thus, the score function can be written as

$$u(\mathbf{x}; \theta) = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} w[n] \mu_\theta[n].$$

The Fisher Information is the expected square of the score:

$$I(\theta) = \mathbb{E} [u(\mathbf{x}; \theta)^2].$$

Hence

$$I(\theta) = \mathbb{E} \left[\left(\frac{1}{\sigma^2} \sum_{n=0}^{N-1} w[n] \mu_\theta[n] \right)^2 \right].$$

Pulling the constant outside gives

$$I(\theta) = \frac{1}{\sigma^4} \mathbb{E} \left[\left(\sum_{n=0}^{N-1} w[n] \mu_\theta[n] \right)^2 \right].$$

Expanding the square,

$$I(\theta) = \frac{1}{\sigma^4} \mathbb{E} \left[\sum_{n=0}^{N-1} \sum_{m=0}^{N-1} w[n] w[m] \mu_\theta[n] \mu_\theta[m] \right].$$

Moving the expectation inside the sums,

$$I(\theta) = \frac{1}{\sigma^4} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \mu_\theta[n] \mu_\theta[m] \mathbb{E}[w[n] w[m]].$$

Because the noise samples are independent, zero-mean, and have variance σ^2 ,

$$\mathbb{E}[w[n] w[m]] = \sigma^2 \delta[n - m],$$

where

$$\delta[n - m] = \begin{cases} 1, & n = m, \\ 0, & n \neq m. \end{cases}$$

Thus

$$I(\theta) = \frac{1}{\sigma^4} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \mu_\theta[n] \mu_\theta[m] \sigma^2 \delta[n - m].$$

The Kronecker delta removes all cross terms where $n \neq m$, leaving

$$I(\theta) = \frac{1}{\sigma^4} \sum_{n=0}^{N-1} \sigma^2 \mu_\theta[n]^2.$$

Therefore

$$I(\theta) = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} \mu_\theta[n]^2.$$

Substituting back

$$\mu_\theta[n] = \frac{\partial \mu[n; \theta]}{\partial \theta}$$

gives

$$I(\theta) = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} \left(\frac{\partial \mu[n; \theta]}{\partial \theta} \right)^2. \quad (26)$$

Key idea

For a deterministic signal in additive Gaussian noise, Fisher Information is proportional to the squared sensitivity of the expected signal to the unknown parameter.

7 Applying this to time-delay estimation

For one echo, let

$$\mu[n; \tau] = As(nT_s - \tau). \quad (27)$$

Then

$$\frac{\partial \mu[n; \tau]}{\partial \tau} = -As'(nT_s - \tau), \quad (28)$$

where $s'(t)$ denotes the time derivative of the transmitted waveform.

Therefore

$$I(\tau) = \frac{A^2}{\sigma^2} \sum_{n=0}^{N-1} [s'(nT_s - \tau)]^2. \quad (29)$$

For sufficiently dense sampling, the sum may be approximated by an integral:

$$\sum_n [s'(nT_s - \tau)]^2 \approx f_s \int |s'(t)|^2 dt. \quad (30)$$

Thus

$$I(\tau) \approx \frac{A^2 f_s}{\sigma^2} \int |s'(t)|^2 dt. \quad (31)$$

Key idea

A waveform gives more timing information when it changes rapidly in time. Slowly varying signals have small derivatives, so their delays are harder to estimate accurately.

8 Why bandwidth matters

Using Parseval's theorem, the derivative energy can be written in the frequency domain. If $S(f)$ is the Fourier transform of $s(t)$, then

$$\int |s'(t)|^2 dt = \int (2\pi f)^2 |S(f)|^2 df. \quad (32)$$

Therefore

$$I(\tau) \approx \frac{A^2 f_s}{\sigma^2} \int (2\pi f)^2 |S(f)|^2 df. \quad (33)$$

This shows why high-frequency and wide-band signals are useful for delay estimation. If the signal has more energy at higher frequencies, then small time shifts produce larger waveform changes.

Define the signal energy

$$E_s = \int |s(t)|^2 dt \quad (34)$$

and the mean-square bandwidth

$$\beta^2 = \frac{\int f^2 |S(f)|^2 df}{\int |S(f)|^2 df}. \quad (35)$$

Then

$$\int (2\pi f)^2 |S(f)|^2 df = 4\pi^2 \beta^2 E_s. \quad (36)$$

Hence

$$I(\tau) \approx \frac{A^2 f_s}{\sigma^2} 4\pi^2 \beta^2 E_s. \quad (37)$$

The delay CRLB is therefore approximately

$$\text{var}(\hat{\tau}) \geq \frac{\sigma^2}{4\pi^2 A^2 f_s \beta^2 E_s}. \quad (38)$$

For distance estimation,

$$\text{var}(\hat{d}) \geq \left(\frac{c}{2}\right)^2 \frac{\sigma^2}{4\pi^2 A^2 f_s \beta^2 E_s}. \quad (39)$$

9 Interpretation for the AIR SONAR parameters

Your real-time sonar code uses parameters such as

```
fs = 48000 # sampling frequency
f0 = 6000 # chirp start frequency
f1 = 12000 # chirp end frequency
Npulse = 500 # chirp duration in samples
Nseg = 2048*2 # samples between pulses
maxdist = 200 # maximum distance in cm
```

The chirp duration is

$$T_p = \frac{N_{\text{pulse}}}{f_s}. \quad (40)$$

For $N_{\text{pulse}} = 500$ and $f_s = 48000$, this is

$$T_p \approx \frac{500}{48000} \approx 10.4 \text{ ms}. \quad (41)$$

The chirp bandwidth is

$$B = f_1 - f_0. \quad (42)$$

For $f_0 = 6000 \text{ Hz}$ and $f_1 = 12000 \text{ Hz}$,

$$B = 6000 \text{ Hz}. \quad (43)$$

A rough matched-filter range resolution is

$$\Delta d \approx \frac{c}{2B}. \quad (44)$$

For $B = 6000 \text{ Hz}$,

$$\Delta d \approx \frac{343}{2 \times 6000} \approx 0.0286 \text{ m} \approx 2.9 \text{ cm}. \quad (45)$$

This is not exactly the same as the CRLB, but it has the same message: larger bandwidth improves distance discrimination.

Example

Increasing the chirp from 6000–12000 Hz to 4000–16000 Hz doubles the bandwidth from 6 kHz to 12 kHz. The rough range resolution improves from about 2.9 cm to about 1.4 cm, assuming the loudspeaker and microphone work well over that band.

10 Trade-offs in choosing sonar parameters

10.1 Pulse length

A longer pulse has larger energy:

$$E_s = \sum_n |s[n]|^2. \quad (46)$$

If the pulse amplitude is fixed, increasing N_{pulse} increases E_s . This increases Fisher Information and improves robustness to noise.

However, a longer non-chirped pulse has poorer time resolution. A long narrowband tone is difficult to localise in time. This is why chirps are useful: they allow long duration and high energy while maintaining good delay resolution after matched filtering.

10.2 Bandwidth

Increasing $f_1 - f_0$ increases bandwidth and usually improves delay estimation. But practical limits matter:

- laptop speakers may not reproduce very low or very high frequencies well;
- microphones may have uneven frequency response;
- high-frequency sound attenuates more strongly in air;
- frequencies near the Nyquist frequency can be problematic.

10.3 Sampling frequency

At $f_s = 48$ kHz, one sample corresponds to

$$T_s = \frac{1}{48000} \approx 20.8 \mu\text{s}. \quad (47)$$

The corresponding one-sample distance spacing is

$$\Delta d_{\text{sample}} = \frac{cT_s}{2} = \frac{c}{2f_s} \approx \frac{343}{96000} \approx 3.6 \text{ mm}. \quad (48)$$

This is the spacing between possible integer-sample delay estimates. In practice, the matched-filter peak width and noise often dominate over this sampling limit.

10.4 Segment length and maximum range

The code uses N_{seg} as the number of samples between pulses. This determines the maximum unambiguous delay. The corresponding maximum distance is approximately

$$d_{\text{max}} = \frac{c}{2} \frac{N_{\text{seg}}}{f_s}. \quad (49)$$

For $N_{\text{seg}} = 4096$ and $f_s = 48000$,

$$d_{\text{max}} \approx \frac{343}{2} \frac{4096}{48000} \approx 14.6 \text{ m}. \quad (50)$$

However, real laptop sonar is usually limited by loudspeaker/microphone strength, room reflections, and background noise long before this theoretical limit.

11 A practical Fisher Information computation in Python

For a discrete pulse $s[n]$, the derivative $s'(t)$ can be approximated by a finite difference:

$$s'(nT_s) \approx f_s(s[n] - s[n-1]). \quad (51)$$

Then a simple numerical approximation to the Fisher Information for delay is

$$I(\tau) \approx \frac{A^2}{\sigma^2} \sum_n (f_s(s[n] - s[n-1]))^2. \quad (52)$$


```

def fisher_delay_information(pulse, fs, A=1.0, sigma=1.0):
    """
    Approximate Fisher Information for estimating the delay
    of a known pulse in additive white Gaussian noise.
    """
    pulse = np.asarray(pulse)
    dpulse_dt = fs * np.diff(pulse, prepend=pulse[0])
    I_tau = (A**2 / sigma**2) * np.sum(np.abs(dpulse_dt)**2)
    return I_tau

def crlb_distance_from_pulse(pulse, fs, c=343.0, A=1.0, sigma=1.0):
    """
    Approximate CRLB for distance estimation from a known pulse.
    """
    I_tau = fisher_delay_information(pulse, fs, A=A, sigma=sigma)
    var_tau = 1.0 / I_tau
    var_d = (c/2)**2 * var_tau
    std_d = np.sqrt(var_d)
    return var_d, std_d

```

This code is deliberately simplified. In real sonar experiments:

- the noise may not be white;
- room reverberation creates multiple echoes;
- the loudspeaker and microphone distort the transmitted chirp;
- the echo amplitude A is unknown and changes with distance and target orientation;
- the direct path from speaker to microphone may dominate the recording.

Nevertheless, the formula gives useful intuition.

12 Unknown amplitude: estimating both amplitude and delay

In practice, the echo amplitude A is usually unknown. A more realistic model is

$$x[n] = As(nT_s - \tau) + w[n]. \quad (53)$$

Now the unknown parameter vector is

$$\boldsymbol{\theta} = \begin{bmatrix} A \\ \tau \end{bmatrix}. \quad (54)$$

The Fisher Information becomes a matrix:

$$\mathbf{I}(\boldsymbol{\theta}) = \frac{1}{\sigma^2} \begin{bmatrix} \sum_n s_n^2 & -A \sum_n s_n s'_n \\ -A \sum_n s_n s'_n & A^2 \sum_n (s'_n)^2 \end{bmatrix}, \quad (55)$$

where

$$s_n = s(nT_s - \tau), \quad s'_n = s'(nT_s - \tau). \quad (56)$$

The CRLB for τ is obtained from the inverse matrix:

$$\text{var}(\hat{\tau}) \geq [\mathbf{I}^{-1}(\boldsymbol{\theta})]_{2,2}. \quad (57)$$

Note the index (2,2) refers to the element in row 2, column 2 of the matrix.

If the signal is well-localised and boundary effects are small, then

$$\sum_n s_n s'_n \approx 0, \quad (58)$$

so the amplitude and delay estimates are nearly decoupled. In that case we approximately recover

$$\text{var}(\hat{\tau}) \gtrsim \frac{\sigma^2}{A^2 \sum_n (s'_n)^2}. \quad (59)$$

13 Matrix CRLB and the Geometry of Ellipses

13.1 From scalar to vector CRLB

For a scalar parameter,

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)}.$$

For a vector parameter $\boldsymbol{\theta} \in \mathbb{R}^d$,

$$\text{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{I}^{-1}(\boldsymbol{\theta}),$$

which means

$$\mathbf{z}^T (\text{Cov}(\hat{\boldsymbol{\theta}}) - \mathbf{I}^{-1}(\boldsymbol{\theta})) \mathbf{z} \geq 0 \quad \forall \mathbf{z} \in \mathbb{R}^d.$$

13.2 Covariance matrices define quadratic forms

Let $\boldsymbol{\Sigma} = \text{Cov}(\hat{\boldsymbol{\theta}})$. Consider the quadratic form

$$Q(\mathbf{e}) = \mathbf{e}^T \boldsymbol{\Sigma}^{-1} \mathbf{e}, \quad \mathbf{e} = \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}.$$

The set of points satisfying

$$\mathbf{e}^T \boldsymbol{\Sigma}^{-1} \mathbf{e} = c$$

is an *ellipse* in $\mathbb{R}^2$ (and an ellipsoid in higher dimensions).

Why an ellipse? Because $\boldsymbol{\Sigma}^{-1}$ is symmetric positive definite (for non-degenerate covariance), it admits an eigen-decomposition

$$\boldsymbol{\Sigma}^{-1} = \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}^T,$$

where $\mathbf{U}$ is orthonormal and $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_d)$ with $\lambda_i > 0$.

Define rotated coordinates $\mathbf{y} = \mathbf{U}^T \mathbf{e}$. Then

$$\mathbf{e}^T \boldsymbol{\Sigma}^{-1} \mathbf{e} = \mathbf{y}^T \boldsymbol{\Lambda} \mathbf{y} = \sum_{i=1}^d \lambda_i y_i^2.$$

Thus the level set

$$\sum_{i=1}^d \lambda_i y_i^2 = c$$

is an axis-aligned ellipse (ellipsoid), with principal axes along the eigenvectors of $\boldsymbol{\Sigma}$ and axis lengths proportional to $1/\sqrt{\lambda_i}$.

13.3 Statistical interpretation: uncertainty ellipses

If $\hat{\boldsymbol{\theta}}$ is approximately Gaussian,

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}(\boldsymbol{\theta}, \boldsymbol{\Sigma}),$$

then contours of equal density satisfy

$$(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})^T \boldsymbol{\Sigma}^{-1} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) = c,$$

which are ellipses (ellipsoids). These are called *confidence ellipses*.

13.4 What does the matrix inequality mean geometrically?

The CRLB states:

$$\mathbf{\Sigma} \succeq \mathbf{I}^{-1}.$$

Equivalently,

$$\mathbf{\Sigma}^{-1} \preceq \mathbf{I}.$$

Consider the corresponding quadratic forms:

$$\mathbf{e}^T \mathbf{\Sigma}^{-1} \mathbf{e} \leq \mathbf{e}^T \mathbf{I}(\boldsymbol{\theta}) \mathbf{e} \quad \forall \mathbf{e}.$$

This implies:

The uncertainty ellipse defined by $\mathbf{\Sigma}$ is *never smaller* than the ellipse defined by $\mathbf{I}^{-1}$.

13.5 Ellipse containment interpretation

Define two ellipsoids:

$$\mathcal{E}_{\Sigma} = \{\mathbf{e} : \mathbf{e}^T \mathbf{\Sigma}^{-1} \mathbf{e} \leq 1\},$$

$$\mathcal{E}_{\text{CRLB}} = \{\mathbf{e} : \mathbf{e}^T \mathbf{I}(\boldsymbol{\theta}) \mathbf{e} \leq 1\}.$$

Then

$$\mathbf{\Sigma} \succeq \mathbf{I}^{-1} \iff \mathcal{E}_{\text{CRLB}} \subseteq \mathcal{E}_{\Sigma}.$$

Interpretation: The CRLB ellipsoid is the *smallest achievable uncertainty region*. Any unbiased estimator must have an uncertainty region that contains it.

13.6 Directional interpretation

For any unit vector $\mathbf{z}$, consider the variance in direction $\mathbf{z}$:

$$\text{Var}(\mathbf{z}^T \hat{\boldsymbol{\theta}}) = \mathbf{z}^T \mathbf{\Sigma} \mathbf{z}.$$

The CRLB implies

$$\mathbf{z}^T \mathbf{\Sigma} \mathbf{z} \geq \mathbf{z}^T \mathbf{I}^{-1} \mathbf{z}.$$

In *every direction*, the variance is bounded below by the corresponding directional CRLB.

13.7 Two-parameter example

Let

$$\boldsymbol{\theta} = \begin{bmatrix} A \\ \tau \end{bmatrix}$$

(e.g. amplitude and delay).

Then

$$\mathbf{I}(\boldsymbol{\theta}) = \begin{bmatrix} I_{AA} & I_{A\tau} \\ I_{\tau A} & I_{\tau\tau} \end{bmatrix}.$$

The inverse is

$$\mathbf{I}^{-1} = \frac{1}{I_{AA}I_{\tau\tau} - I_{A\tau}^2} \begin{bmatrix} I_{\tau\tau} & -I_{A\tau} \\ -I_{\tau A} & I_{AA} \end{bmatrix}.$$

Thus

$$\text{Var}(\hat{\tau}) \geq [\mathbf{I}^{-1}]_{2,2} = \frac{I_{AA}}{I_{AA}I_{\tau\tau} - I_{A\tau}^2}.$$

Key insight: The achievable accuracy for τ depends on *all* entries of the Fisher Information Matrix, FIM, not just $I_{\tau\tau}$. Parameter coupling (nonzero off-diagonal terms) enlarges the uncertainty ellipse.

13.8 Summary

- A covariance matrix defines ellipses (ellipsoids) via quadratic forms
- The matrix CRLB compares these ellipsoids
- The CRLB ellipsoid is the smallest achievable uncertainty region
- Estimation error cannot be reduced below this bound in any direction

The matrix CRLB is a geometric statement: it bounds the size and shape of the estimation uncertainty ellipsoid.

14 Important conceptual message

The AIR SONAR project is not just a coding exercise. It contains a complete signal processing chain:

waveform design $\rightarrow$ transmission $\rightarrow$ room acoustics $\rightarrow$ recording $\rightarrow$ matched filtering $\rightarrow$ delay estimation $\rightarrow$ distance estimation (60)

Fisher Information sits underneath this chain. It explains why some choices of transmitted waveform make distance estimation easier than others.

Parameter change	Effect on Fisher Information	Practical interpretation
Increase signal amplitude A	Increases as A^2	louder echo improves accuracy
Decrease noise variance σ^2	Increases as $1/\sigma^2$	quieter recordings improve accuracy
Increase pulse energy E_s	Increases approximately linearly	longer pulses improve robustness
Increase bandwidth β	Increases as β^2	wider chirps improve timing accuracy
Increase sample rate f_s	Often improves numerical resolution	finer delay grid, better derivative estimation

15 Summary

Fisher Information provides a principled way to understand distance estimation in the AIR SONAR project. The central result is that delay-estimation accuracy improves when the received signal is strong, the noise is low, the pulse has high energy, and the waveform has large effective bandwidth.

For the sonar project:

$$r(t) = As(t - \tau) + w(t), \quad d = \frac{c\tau}{2}. \quad (61)$$

The approximate Fisher Information for delay is

$$I(\tau) \approx \frac{A^2 f_s}{\sigma^2} \int |s'(t)|^2 dt. \quad (62)$$

The corresponding distance CRLB is

$$\text{var}(\hat{d}) \geq \left(\frac{c}{2}\right)^2 \frac{1}{I(\tau)}. \quad (63)$$

Key idea

In practical terms: a good sonar waveform is one whose delay is easy to detect. Fisher Information says this happens when the waveform changes rapidly, has enough energy, and survives the real loudspeaker–room–microphone system.